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BM4152 Biosignal Processing

Assignment 03

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1. Continuous Wavelet Transform

1.1. Introduction

In this section, the normalized Mexican-hat wavelet is derived and its properties are verified. The wavelet is then used in the CWT to analyse the time-varying frequency content of a non-stationary signal.

1.2. Wavelet properties

(i) Derivation of the Mexican-hat function

With $\mu = 0$ and $\sigma = 1$ the Gaussian is

$$g(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}. \quad (1)$$

Differentiating twice,

$$\frac{d}{dt}g(t) = -t g(t), \quad (2)$$

$$\frac{d^2}{dt^2}g(t) = -g(t) + t^2 g(t) = (t^2 - 1) g(t), \quad (3)$$

so that

$$m(t) = -\frac{d^2}{dt^2}g(t) = \frac{1}{\sqrt{2\pi}} (1 - t^2) e^{-t^2/2}. \quad (4)$$

(ii) Normalizing factor

The energy of $m(t)$ is

$$E = \int_{-\infty}^{\infty} m^2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} (1 - 2t^2 + t^4) e^{-t^2} dt. \quad (5)$$

Using the standard Gaussian integrals $\int e^{-t^2} dt = \sqrt{\pi}$, $\int t^2 e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$ and $\int t^4 e^{-t^2} dt = \frac{3\sqrt{\pi}}{4}$,

$$E = \frac{1}{2\pi} \left(\sqrt{\pi} - \sqrt{\pi} + \frac{3\sqrt{\pi}}{4} \right) = \frac{3}{8\sqrt{\pi}} \approx 0.2116. \quad (6)$$

For unit energy, $m(t)$ must be multiplied by

$$c = \frac{1}{\sqrt{E}} = \sqrt{\frac{8\sqrt{\pi}}{3}} \approx 2.174. \quad (7)$$

(iii) Normalized mother and daughter wavelets

$$\psi(t) = c m(t) = \frac{2}{\sqrt{3} \pi^{1/4}} (1 - t^2) e^{-t^2/2}, \quad \frac{2}{\sqrt{3} \pi^{1/4}} \approx 0.8673. \quad (8)$$

Including the scale s and the translation τ , the generic daughter wavelet is

$$\psi_{s,\tau}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right) = \frac{2}{\sqrt{3s} \pi^{1/4}} \left[1 - \left(\frac{t-\tau}{s}\right)^2\right] \exp\left[-\frac{1}{2} \left(\frac{t-\tau}{s}\right)^2\right]. \quad (9)$$

The factor $1/\sqrt{s}$ keeps the energy of every daughter wavelet equal to that of the mother wavelet.

(iv) Time-domain daughter wavelets

The function `mexican_hat(t, scale, translation)` of the provided script was completed with the expression above and evaluated for the 20 scales $s = 0.01, 0.11, \dots, 1.91$ on the time axis $t = n/f_s$, $n = -3000, \dots, 3000$, $f_s = 250$ Hz. The figures 1 and 2 show the corresponding time-domain daughter wavelets.

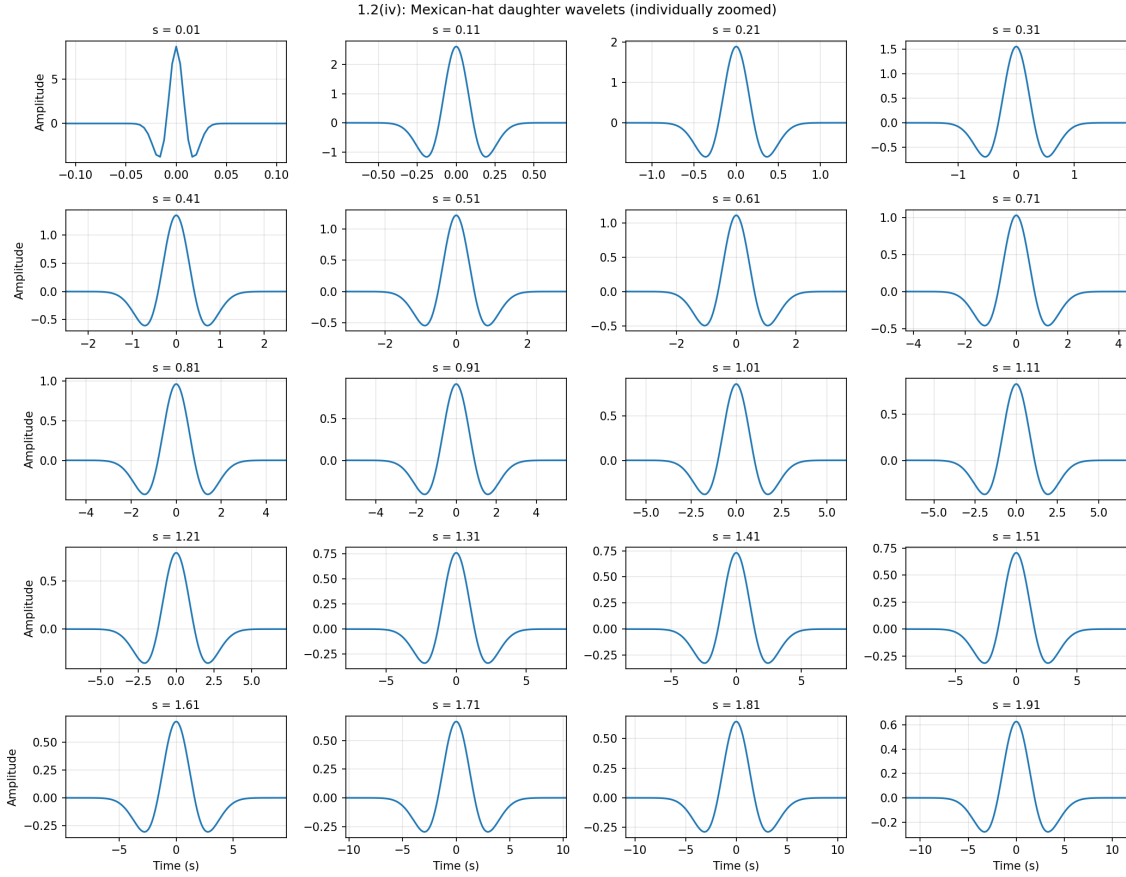


Figure 1: Mexican-hat daughter wavelets for $s = 0.01 + 0.1k$, $k = 0, \dots, 19$ (each panel zoomed to $\pm 6s$).

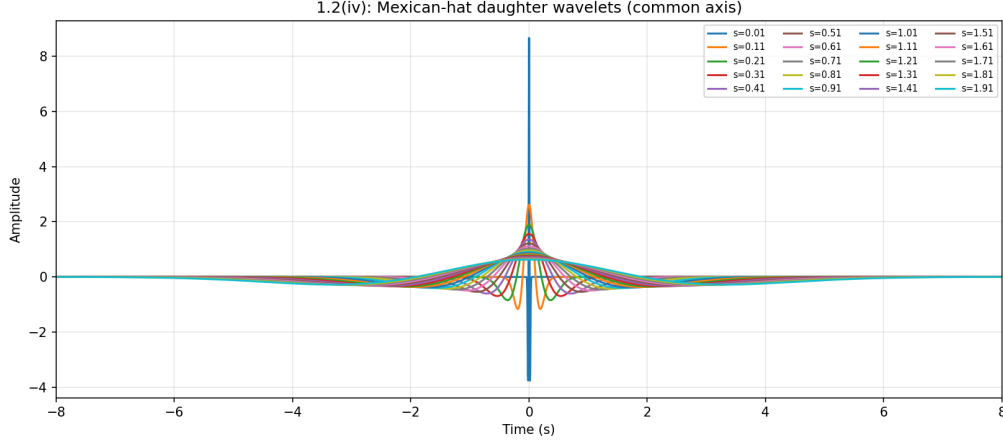


Figure 2: The same daughter wavelets on a common time axis.

All daughter wavelets have the same shape. As the scale increases the wavelet is stretched in time and, because of the $1/\sqrt{s}$ factor, its peak amplitude decreases as $0.8673/\sqrt{s}$. Small scales give narrow, tall wavelets and large scales give wide, low wavelets.

(v) Verification of wavelet properties

- **Zero mean:** The mean was calculated numerically for all scales. The values are very close to zero, confirming the zero mean property of the daughter wavelets.
- **Unit energy:** The energy was also calculated numerically for all scales. The energy is 1.000000 for every scale, confirming the unit energy normalization.
- **Compact support:** By observation, the wavelets are concentrated around their centre and decay rapidly towards zero away from it. However, the Mexican-hat wavelet does not have true compact support because its Gaussian envelope never becomes exactly zero. It therefore has an effective support where the values become negligibly small.

s	Mean	Energy	s	Mean	Energy
0.01	-2.5×10^{-17}	1.000000	1.01	6.8×10^{-17}	1.000000
0.11	6.2×10^{-18}	1.000000	1.11	-4.9×10^{-18}	1.000000
0.21	-3.9×10^{-17}	1.000000	1.21	-1.1×10^{-17}	1.000000
0.31	-4.7×10^{-17}	1.000000	1.31	1.1×10^{-16}	1.000000
0.41	2.4×10^{-17}	1.000000	1.41	3.2×10^{-15}	1.000000
0.51	5.0×10^{-17}	1.000000	1.51	3.2×10^{-13}	1.000000
0.61	-3.4×10^{-17}	1.000000	1.61	1.4×10^{-11}	1.000000
0.71	2.2×10^{-18}	1.000000	1.71	3.2×10^{-10}	1.000000
0.81	5.7×10^{-17}	1.000000	1.81	4.4×10^{-9}	1.000000
0.91	2.8×10^{-17}	1.000000	1.91	4.0×10^{-8}	1.000000

Table 1: Numerically calculated mean and energy of the daughter wavelets.

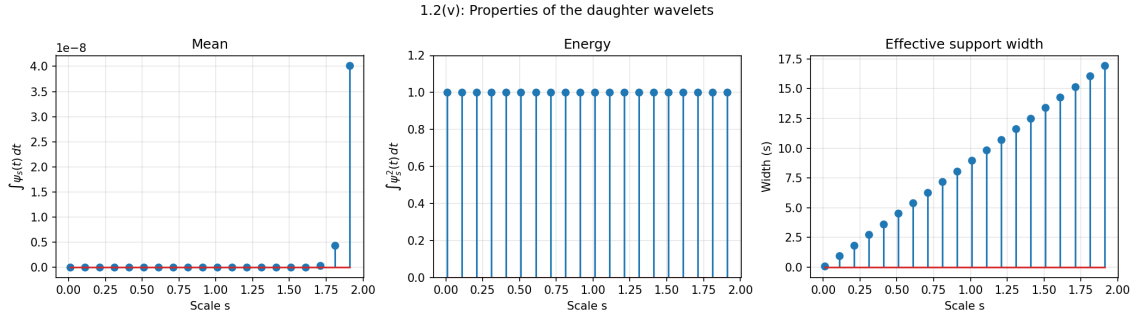


Figure 3: Mean, energy and effective support width of the daughter wavelets against scale.

(vi) Spectra of the daughter wavelets

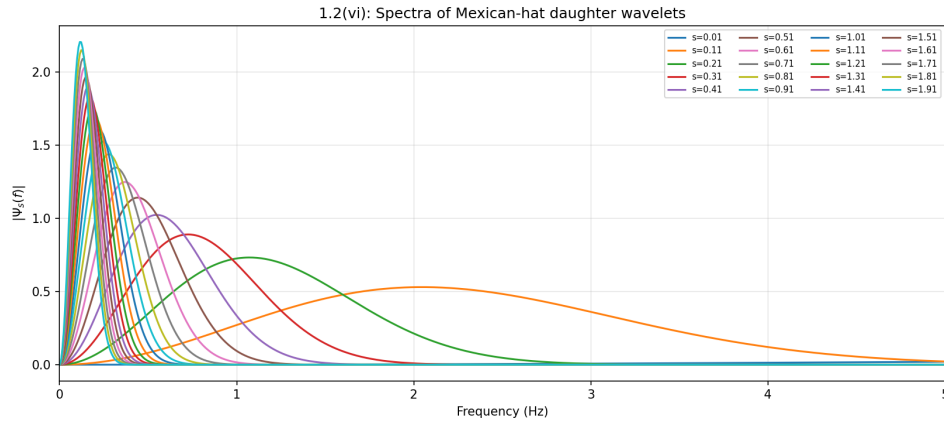


Figure 4: Magnitude spectra of the daughter wavelets (linear frequency axis).

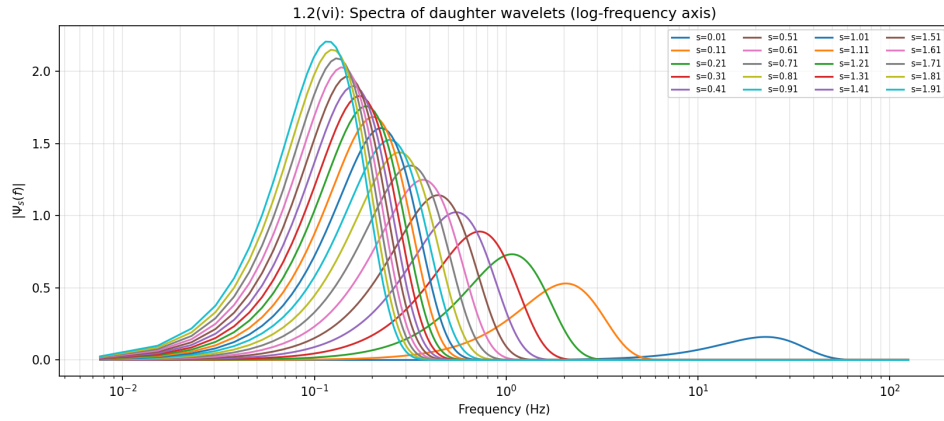


Figure 5: Magnitude spectra of the daughter wavelets on a logarithmic frequency axis.

The spectra show that the frequency content changes with the scale. Smaller scales correspond to higher frequencies, while larger scales correspond to lower frequencies. The peak of the spectrum therefore moves towards lower frequencies as the scale increases. This shows the inverse relationship between scale and frequency and is the reason different scales can be used to detect different frequency components in a signal.

1.3. Continuous wavelet decomposition

(i) Test signal

The signal $x[n]$ was created with $N = 3000$ and $f_s = 250$ Hz. It contains a 0.25 Hz sinusoid for $1 \leq n < 4500$ and a 0.75 Hz sinusoid for $4500 \leq n < 9000$ (Figure 6).

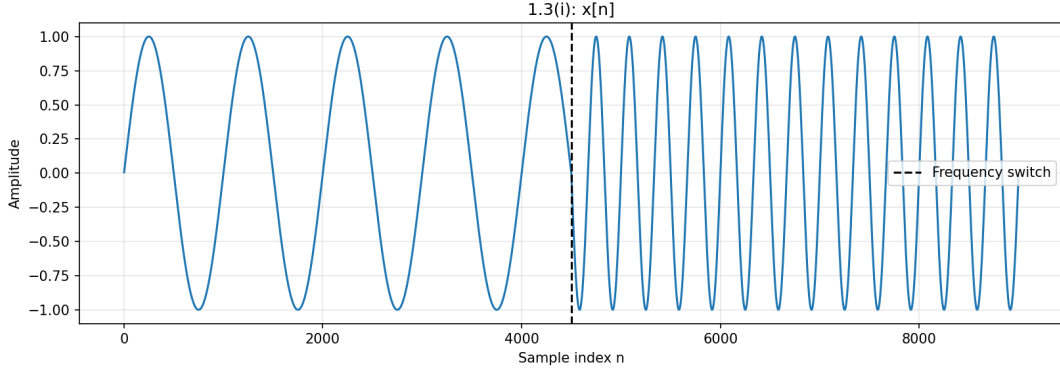


Figure 6: Non-stationary test signal $x[n]$.

(ii) CWT calculation

The CWT was calculated for 200 scales, $s = 0.01, 0.02, \dots, 2.00$. For each scale, the corresponding Mexican-hat daughter wavelet was generated and convolved with the signal using `scipy.signal.fftconvolve(x, wavelet, mode='same')`.

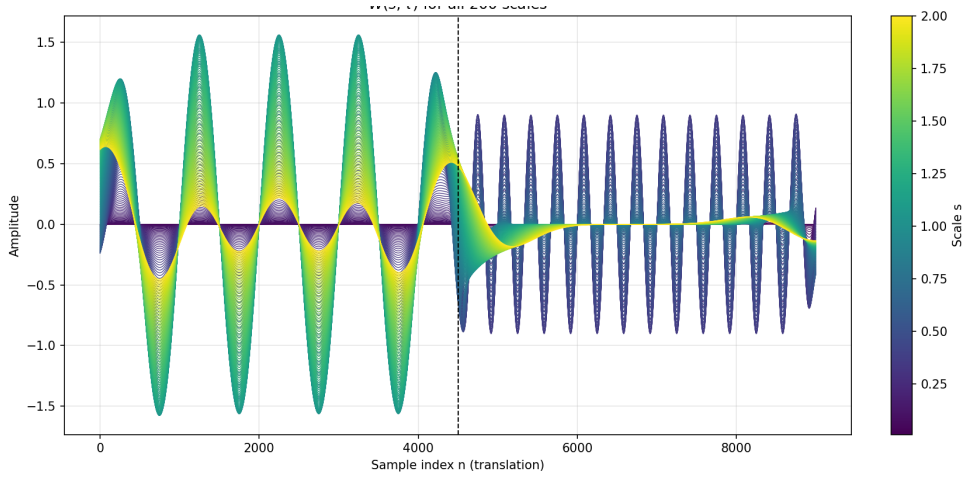


Figure 7: Mexican hat wavelet applied to the input waveform using convolution.

(iii) CWT spectrogram

Repeating the CWT calculation for all 200 scales produced a 200×8999 matrix of CWT coefficients. The magnitude of this coefficient matrix was visualized using `pcolormesh` to show how the signal energy varies with time and scale.

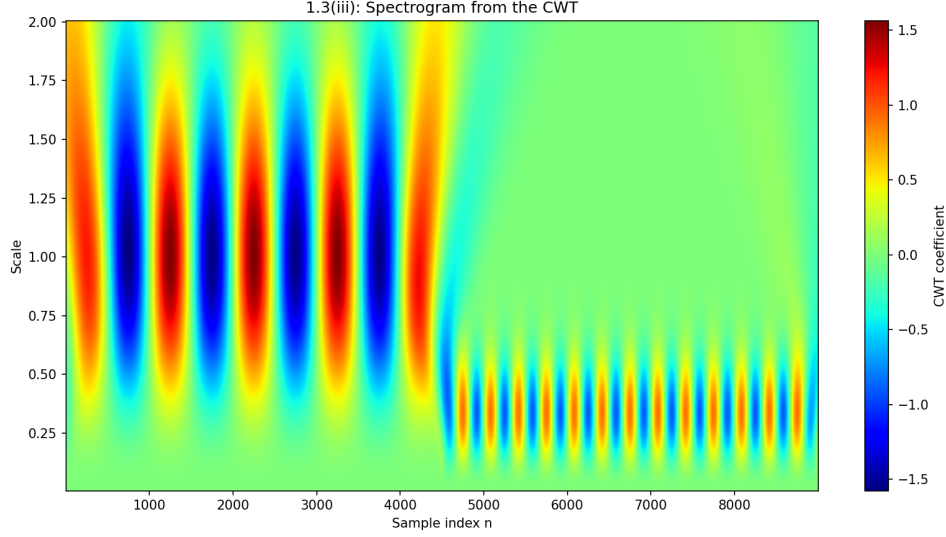


Figure 8: CWT spectrogram of $x[n]$.

(iv) Discussion

The CWT coefficients show how the frequency content of the signal changes with time. In the first half of the signal, the largest coefficients occur around $s \approx 1.0$, corresponding to the lower-frequency component of 0.25 Hz. In the second half, the largest coefficients occur around $s \approx 0.34$, corresponding to the higher-frequency component of 0.75 Hz. Therefore, the change in frequency can be clearly observed around $n = 4500$.

The results also show the relationship between scale and frequency. Larger scales correspond to lower frequencies, while smaller scales correspond to higher frequencies. Since the CWT provides coefficients for different scales at each time position, it shows both **when** a frequency occurs and **which frequency** is present at that time. This allows the time-varying frequency content of $x[n]$ to be identified from the CWT spectrogram.

2. Discrete Wavelet Transform

2.1. Introduction

In this section, the DWT is applied with the db9 and Haar wavelets using PyWavelets, and is then used to denoise two synthetic signals and to compress an ECG recording.

2.2. Applying the DWT with PyWavelets

(i) Test signals

The signals $x_1[n]$ and $x_2[n]$ were created with $f_s = 512$ Hz and 1024 samples. Both were corrupted with additive white Gaussian noise at 10 dB SNR using the provided `awgn` helper with a fixed random seed, giving $y_1[n]$ and $y_2[n]$ (Figure 9).

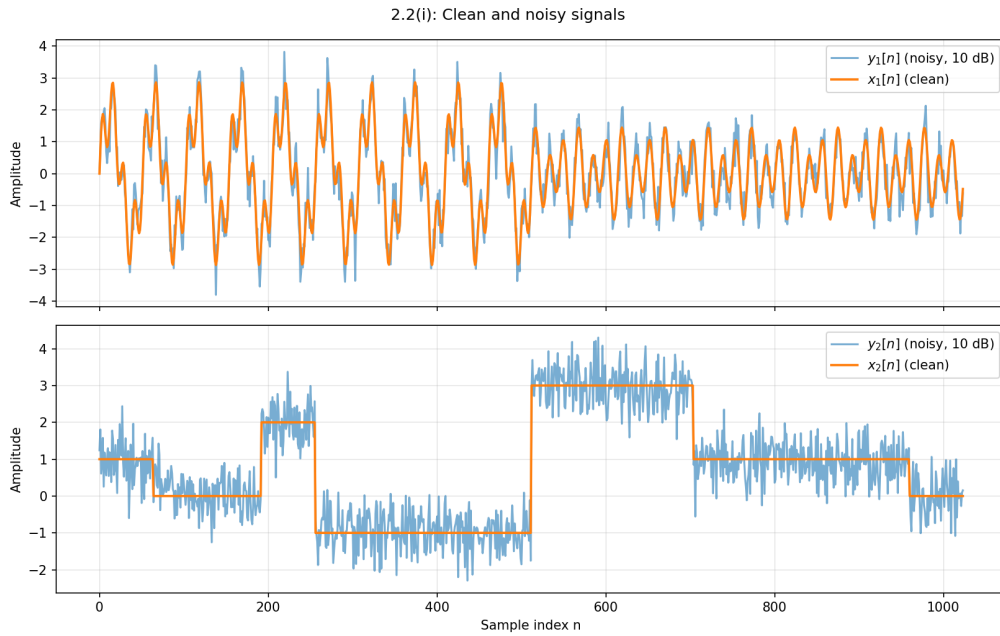


Figure 9: Clean signals $x_1[n]$, $x_2[n]$ and their noisy versions $y_1[n]$, $y_2[n]$ (10 dB SNR).

(ii) Wavelet and scaling functions

Figure 10 shows the scaling function $\phi(t)$ and wavelet function $\psi(t)$ of Daubechies-9 and Haar, obtained with `wavefun()`.

- **Haar:** The scaling function is a rectangular pulse on $[0, 1)$, while the wavelet has a simple positive and negative step shape. Both functions are discontinuous and have very short support. This makes the Haar wavelet well localized in time and suitable for signals with sudden changes, but its frequency representation is less precise.
- **db9:** The scaling and wavelet functions are smoother and more oscillatory than Haar. They have a longer support and nine vanishing moments. Therefore, db9 can represent

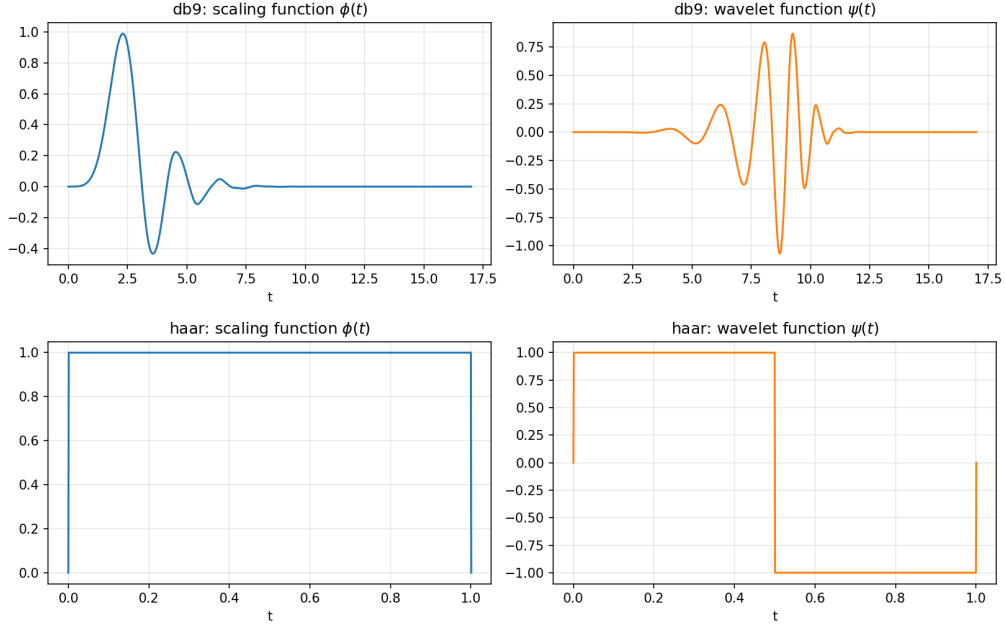


Figure 10: Scaling function $\phi(t)$ and wavelet function $\psi(t)$ of Daubechies-9 and Haar, obtained with `wavefun()`.

smooth signal variations more effectively and provides better frequency resolution, although it is less localized in time than Haar.

(iii) 10-level decomposition

A 10-level DWT was applied to the two noisy signals using the `db9` and `haar` wavelets. The decomposition gives one approximation coefficient set, cA_{10} , and ten detail coefficient sets, $cD_{10}, \dots, cD_1$. Figures 11 and 12 show the resulting wavelet coefficients for both signals and both wavelets.

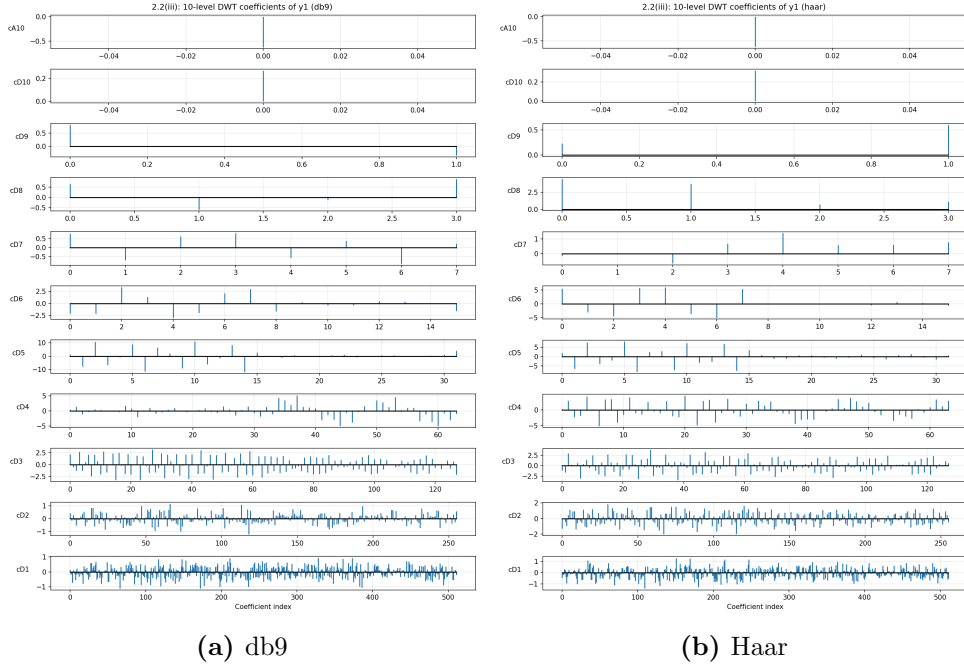


Figure 11: 10-level DWT coefficients $cA_{10}, cD_{10}, \dots, cD_1$ of y_1 .

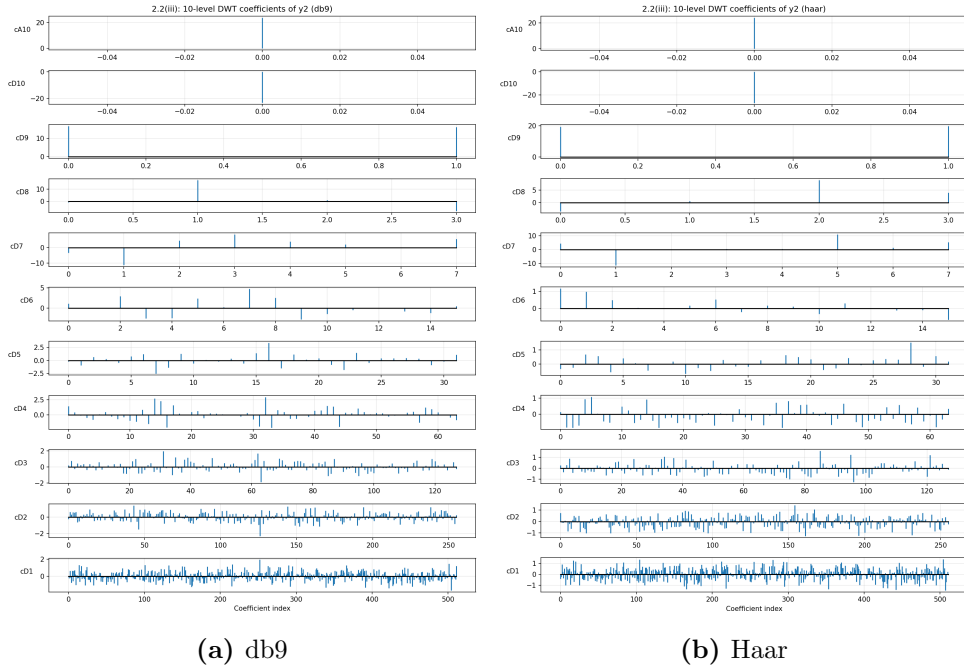


Figure 12: 10-level DWT coefficients $cA_{10}, cD_{10}, \dots, cD_1$ of y_2 .

(iv) Reconstruction of $A_{10}, D_{10}, \dots, D_1$

The steps followed were:

1. Decompose y into the coefficient list $[cA_{10}, cD_{10}, \dots, cD_1]$.

2. To reconstruct one sub-band, keep the corresponding coefficient array and replace all the other arrays with zeros, then apply the inverse DWT (`pywt.waverec`). This gives the contribution of that sub-band alone at the full signal length. Repeating this for every array gives $A_{10}, D_{10}, \dots, D_1$.
3. Add the components, $\hat{y} = A_{10} + \sum_{i=1}^{10} D_i$. Because the inverse DWT is linear, this sum must equal the inverse transform of the complete coefficient set, i.e. the original signal.
4. Compare the energy of y and $\hat{y}$ and compute the energy of the difference, $\sum_n (y[n] - \hat{y}[n])^2$.

Signal	Wavelet	$E(y)$	$E(\hat{y})$	$E(y - \hat{y})$
y_1	db9	1755.721985	1755.721985	2.39×10^{-28}
y_1	Haar	1755.721985	1755.721985	7.48×10^{-28}
y_2	db9	2745.187532	2745.187532	9.71×10^{-28}
y_2	Haar	2745.187532	2745.187532	3.64×10^{-27}

Table 2: Energy of the noisy signal, of the sum of the reconstructed sub-bands and of their difference.

The energies are identical and the energy of the difference is of the order of 10^{-27} . Hence $y = A_{10} + \sum_{i=1}^{10} D_i$ is verified for both signals and both wavelets (perfect reconstruction). The reconstructed sub-bands are shown in Figures 13 and 14.

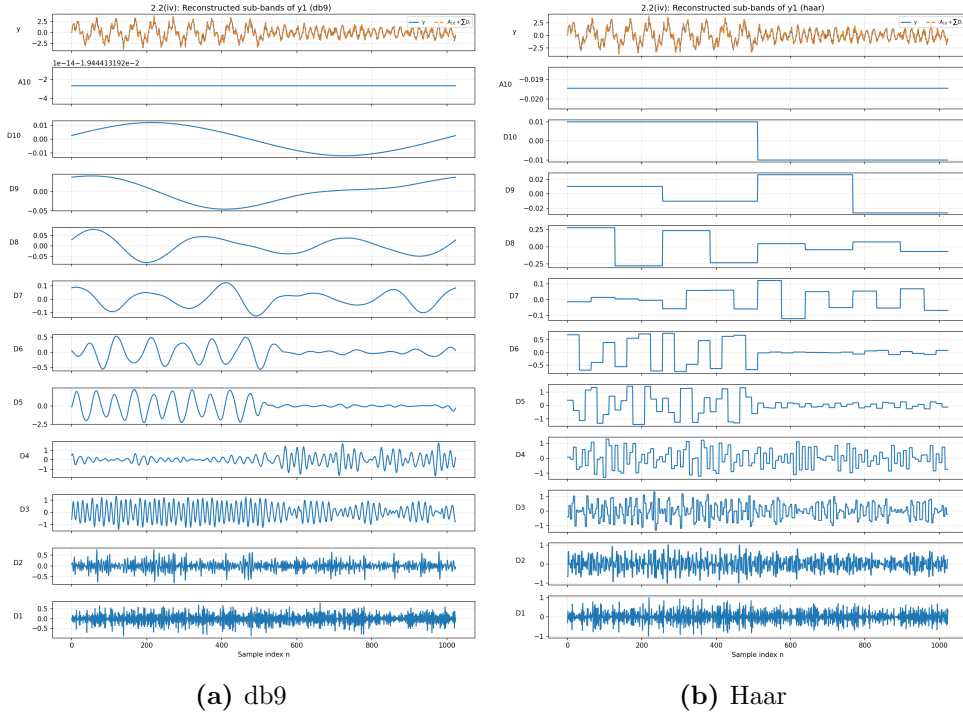


Figure 13: Reconstructed sub-bands $A_{10}, D_{10}, \dots, D_1$ of y_1 .

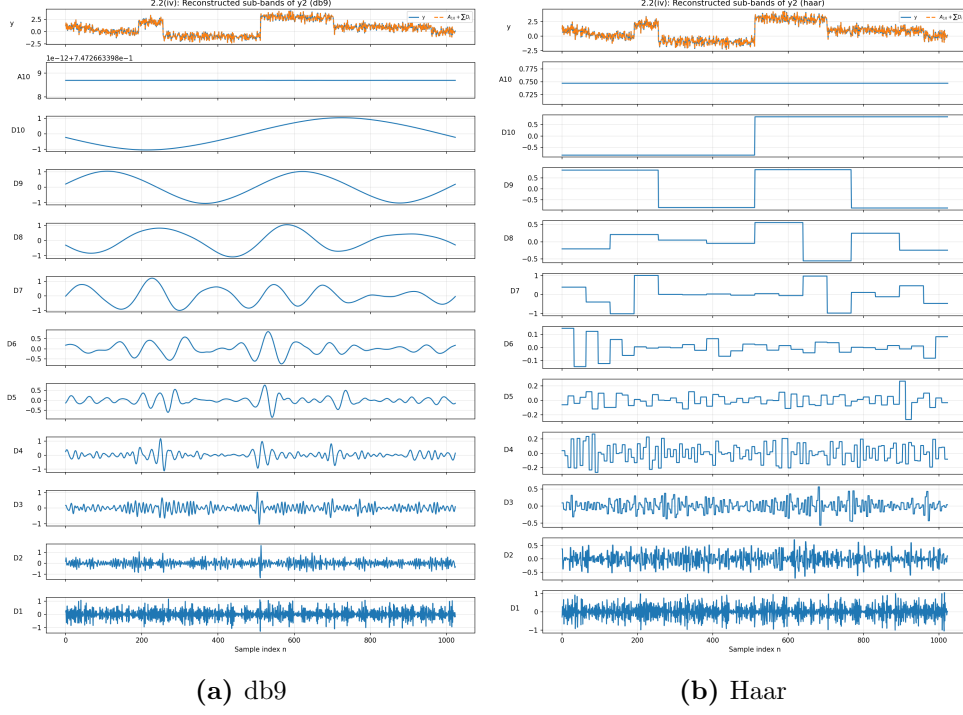


Figure 14: Reconstructed sub-bands $A_{10}, D_{10}, \dots, D_1$ of y_2 .

For y_1 with db9 the sub-bands separate the sinusoids cleanly: the 10 Hz component appears in D_5 (8–16 Hz) and the 40 Hz component in D_3 (32–64 Hz) during the first half, while the 20 Hz and 30 Hz components appear mainly in D_4 (16–32 Hz) during the second half. D_1 and D_2 contain almost only noise. The DWT therefore shows both which frequency bands are active and when they are active.

2.3. Signal denoising with the DWT

(i) Sorted coefficient magnitudes

All the coefficients of each decomposition were concatenated and their magnitudes sorted in descending order (Figure 15).

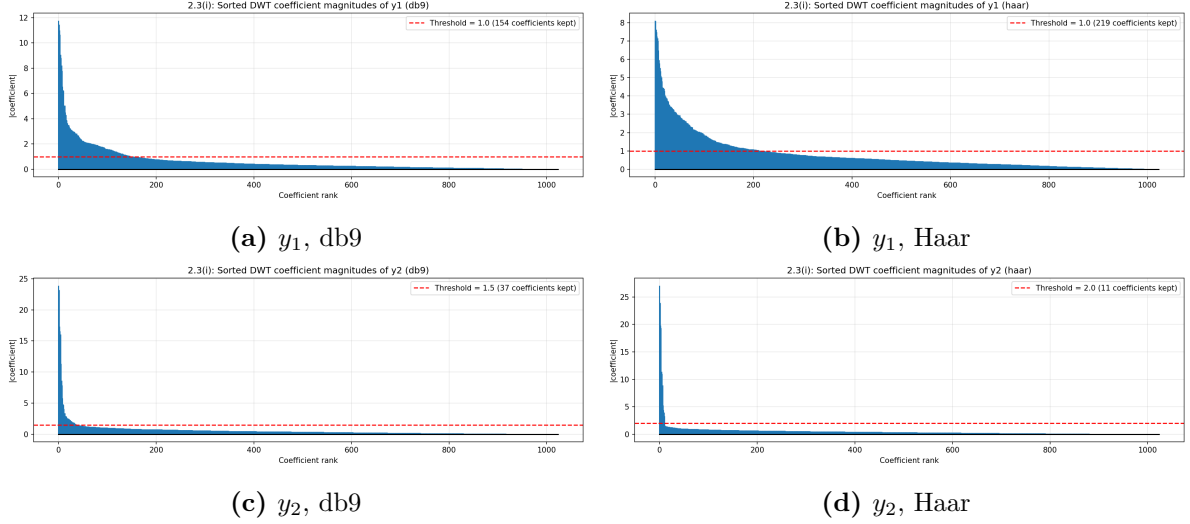


Figure 15: DWT coefficient magnitudes in descending order with the selected thresholds.

(ii) Threshold selection and reconstruction

The threshold was selected by observing the knee of each curve, where the steep decrease changes into a flatter noise floor. The selected thresholds were 1.0 for y_1 with both wavelets, 1.5 for y_2 with db9, and 2.0 for y_2 with Haar. Coefficients with magnitudes below the selected threshold were set to zero using hard thresholding, and the signal was then reconstructed using the inverse DWT.

(iii)–(iv) RMSE and denoised signals

The RMSE between the original clean signal and the denoised signal was calculated as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=1}^N (x[n] - \hat{x}[n])^2},$$

where $x[n]$ is the original signal and $\hat{x}[n]$ is the denoised signal.

Signal	Wavelet	Threshold	Coefficients kept	RMSE (denoised)
x_1	db9	1.0	154 / 1024	0.2532
x_1	Haar	1.0	219 / 1024	0.4160
x_2	db9	1.5	37 / 1024	0.2809
x_2	Haar	2.0	11 / 1024	0.0635

Table 3: RMSE between the original and denoised signals.

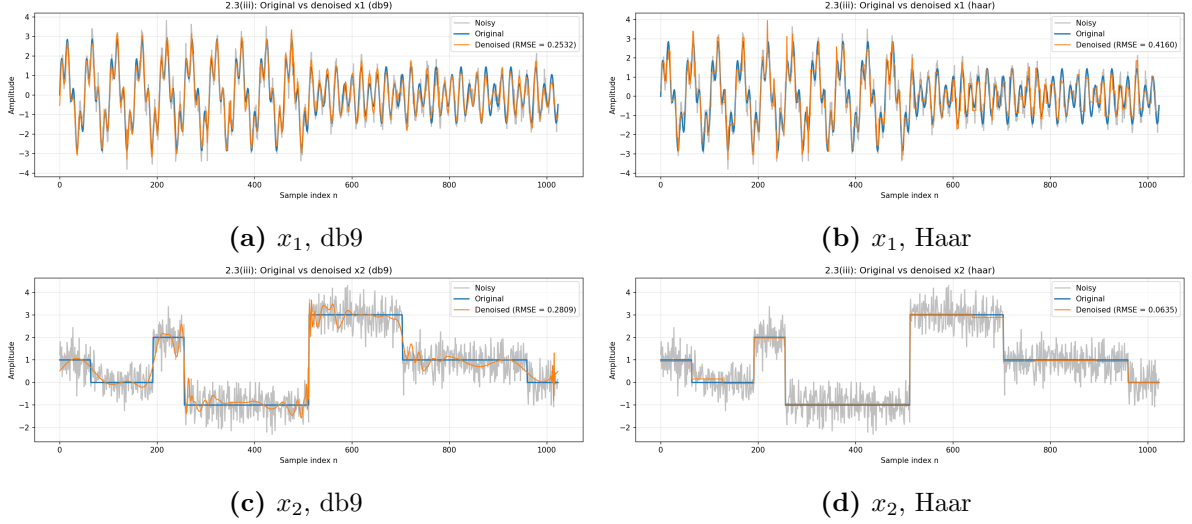


Figure 16: Original, noisy and denoised signals.

For x_1 , the db9 wavelet gives an RMSE of 0.2532 and the denoised signal follows the original sinusoidal waveform closely. The Haar wavelet gives a higher RMSE of 0.4160 and does not preserve the smooth sinusoidal shape as well.

For x_2 , db9 gives an RMSE of 0.2809 and removes much of the noise from the flat regions, but some oscillations can be observed around the step transitions. Haar gives a much lower RMSE of 0.0635 and preserves the step-like shape of the original signal more closely.

(v) Comparison of db9 and Haar

Figure 17 shows the denoised signals obtained with db9 and Haar compared with the originals.

- x_1 (**smooth sinusoids**): db9 is more suitable because it gives a smoother reconstruction and a lower RMSE of 0.2532. Haar produces a less smooth reconstruction and has a higher RMSE of 0.4160.
- x_2 (**piecewise constant**): Haar is more suitable because it preserves the sharp transitions and flat regions of the signal. It gives an RMSE of 0.0635, compared with 0.2809 for db9.

Conclusion: The results show that the choice of wavelet depends on the characteristics of the signal. The smooth db9 wavelet is more suitable for the sinusoidal signal, while the Haar wavelet is more suitable for the piecewise-constant signal with sharp transitions.

2.4. Signal compression with the DWT

(i) Wavelet coefficients of the ECG

The aVR lead (2570 samples, $f_s = 257$ Hz, 10 s) was decomposed with db9 and Haar using the maximum useful level for each wavelet (level 7 for db9 and level 11 for Haar).

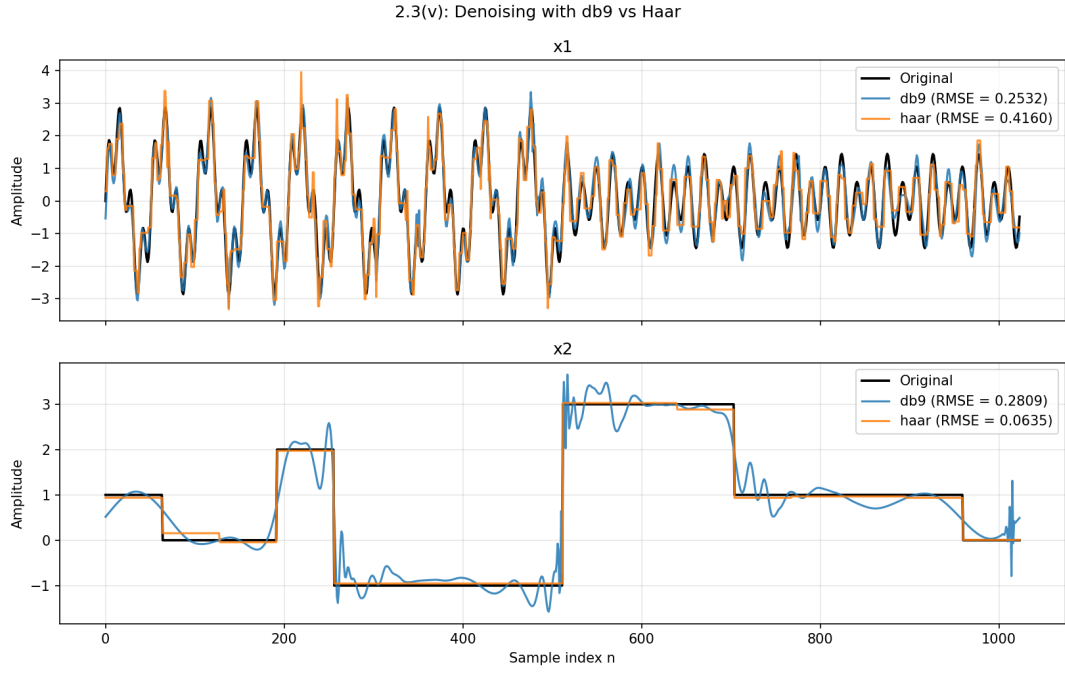


Figure 17: Denoised signals obtained with db9 and Haar compared with the originals.

(ii) Coefficients required for 99% of the energy

The coefficients were sorted by magnitude in descending order and the cumulative sum of their squares was normalized by the total energy. K is the smallest number of coefficients whose cumulative energy reaches 99% (Figures 18 and 19).

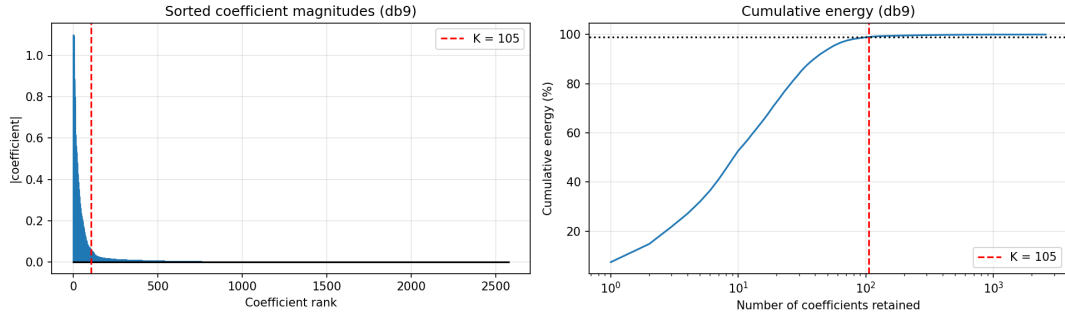


Figure 18: Sorted coefficient magnitudes and cumulative energy of the ECG (db9).

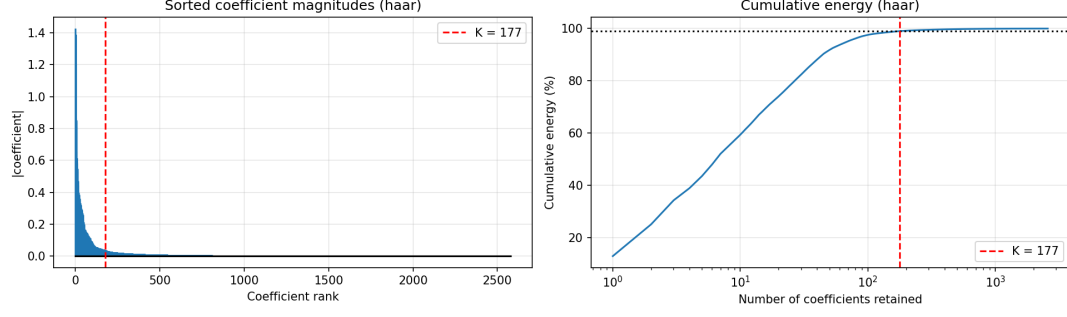


Figure 19: Sorted coefficient magnitudes and cumulative energy of the ECG (Haar).

(iii) Compression

The signal was compressed by keeping the K largest coefficients and setting the rest to zero, and then reconstructed with the inverse DWT. The compression ratio is defined as the number of samples of the original signal divided by the number of retained coefficients, $CR = N/K$.

Wavelet	Level	Total coefficients	K (99% energy)	Compression ratio	RMSE
db9	7	2575	105	24.48 : 1	0.00786
Haar	11	2578	177	14.52 : 1	0.00782

Table 4: ECG compression results at 99% retained energy ($N = 2570$).

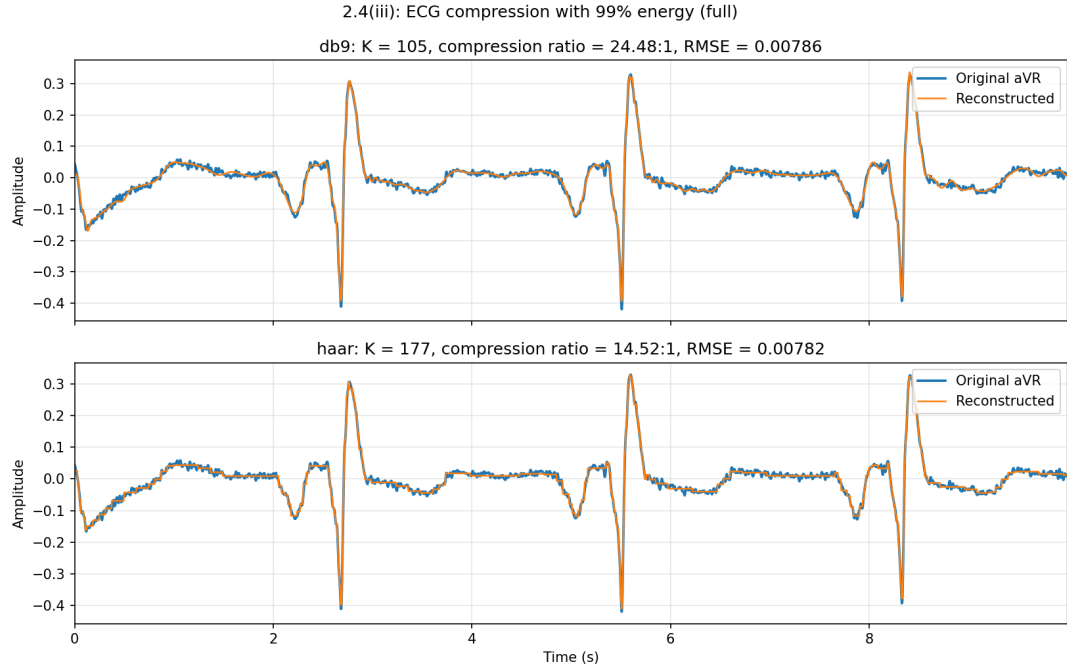


Figure 20: Original and reconstructed ECG (full record).

- **Compression ratio.** About 4% of the coefficients are enough to hold 99% of the energy with db9 (105 coefficients, 24.5:1), while Haar requires 177 coefficients (14.5:1).

The ECG is a mostly smooth signal, so the smooth db9 wavelet with nine vanishing moments represents it more sparsely than the discontinuous Haar wavelet.

- **Morphology with db9.** The reconstructed waveform is smooth and follows the baseline, the P and T waves and the QRS complex closely, with the QRS amplitude and timing preserved. The small high-frequency ripple on the original signal is removed, since it carries little energy; the compression therefore also acts as a denoiser.
- **Morphology with Haar.** The main features and the QRS complex are also preserved, but the slowly varying segments are reconstructed as a staircase (blocking artefacts), which is not physiological and could affect the interpretation of low-amplitude features such as the P wave and the ST segment.

Conclusion. db9 is the more suitable wavelet for ECG compression: it gives a higher compression ratio at the same retained energy and a reconstruction that is visually much closer to a real ECG.